1.2 Innovations and Inventions

In your lifetime, new ideas and technology have transformed communication in dramatic ways. During the early 1800s, new technologies started speeding up transportation and communication.

MAIN IDEA New inventions made American workers more productive, created new industries, and contributed to the nation's economic growth.

NEW MANUFACTURING METHODS

As industries began to flourish in the United States, innovative thinkers were making manufacturing more efficient and profitable. Eli Whitney, John Deere, Robert Fulton, and Samuel Morse were just a few of the many Americans who introduced new technologies in the early 19th century.

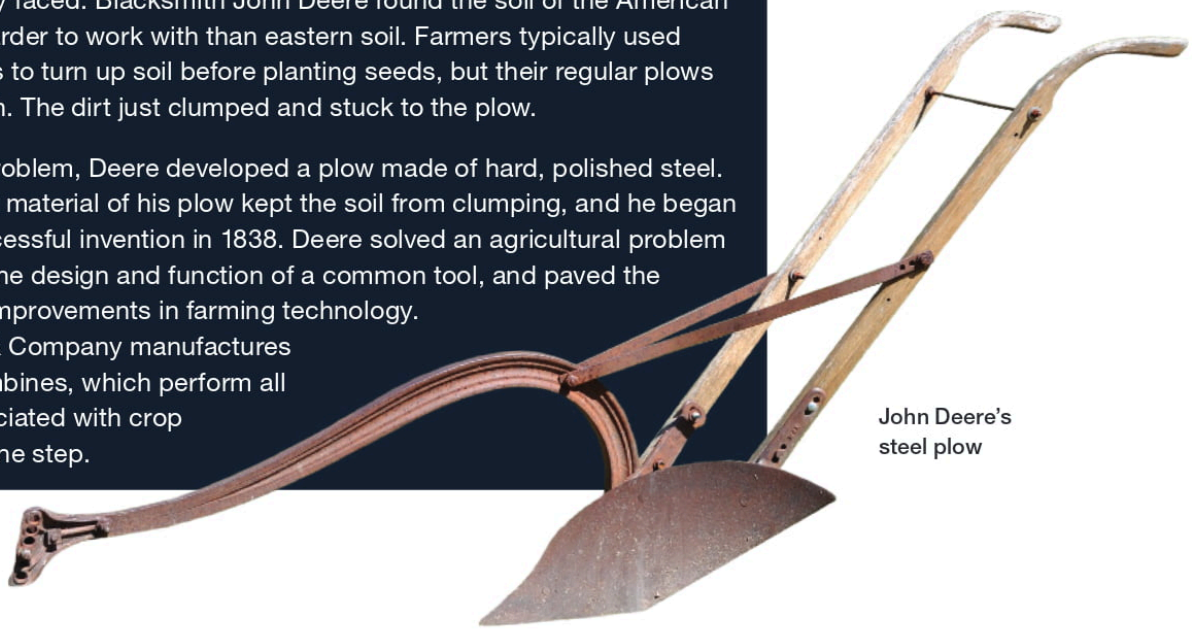
Eli Whitney, one of the most influential of these inventors, was born in 1765 in Massachusetts. From an early age, he showed a natural talent for mechanics and was able to skillfully improve useful items. In 1797, war between the United States and France seemed likely, and the nation's military needed more weapons. Whitney came up with an ambitious plan to supply the federal government with 10,000 rifles in just two years' time. The challenge was this: Every weapon already in use was unique. Any repair required making a new custom part for that specific gun. So Whitney made his rifles using **interchangeable parts**. Each of his guns had the exact same parts.

Whitney completed the government's order in eight years, but his method of using interchangeable parts set the machine age in motion. The use of interchangeable parts soon spread to the production of nearly all goods. Having all parts ready to use meant that goods could be produced quickly and at a lower cost. This efficiency helped shape the identity of the United States as a nation of problem-solving, ingenuity, and progress.

John Deere Historic Site, Grand Detour, Illinois

As Americans moved west, they developed new technologies to tackle the challenges they faced. Blacksmith John Deere found the soil of the American prairie to be harder to work with than eastern soil. Farmers typically used cast-iron plows to turn up soil before planting seeds, but their regular plows weren't enough. The dirt just clumped and stuck to the plow.

To solve this problem, Deere developed a plow made of hard, polished steel. The shape and material of his plow kept the soil from clumping, and he began selling his successful invention in 1838. Deere solved an agricultural problem by examining the design and function of a common tool, and paved the way for more improvements in farming technology. Today, Deere & Company manufactures plows and combines, which perform all the tasks associated with crop harvesting in one step.



John Deere's steel plow

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CRITICAL VIEWING Today's combine harvesters definitely look different from Deere's original plow. These machines accomplish all the harvesting tasks: they thresh, separate, clean, and collect the grain. What effect do you think these machines have on farm life and food production?

STEAMBOATS AND TELEGRAPHS

New technologies also changed shipping, communication, and agriculture. Robert Fulton's **steamboats**, or boats outfitted with steam boiler engines to power their paddle wheels, revolutionized river travel and made shipping quicker. By 1811, Fulton had designed a steamboat powerful and sturdy enough for the strong currents of the Mississippi River. Soon, an upstream journey on the Mississippi and Ohio rivers from New Orleans to Louisville, Kentucky was shortened from several months to just 14 days. But even the swiftest ship could not carry a message across miles in a matter of minutes.

Samuel F. B. Morse's **telegraph** sent messages over electrical wires using a series of long and short pulses known as Morse code. An operator in one location tapped out a message on a device called a key. The message moved through wires to a machine that printed it out as dots and dashes punched onto a strip of paper. Another operator then transcribed the code into words. The first telegraph message was transmitted on May 24, 1844. Soon, telegraph wires stretched throughout the nation, and operators could interpret the code in real time. Widespread use of the telegraph for personal, business, and government communication continued for more than 100 years.

Innovations changed life on the farm as well. Cyrus Hall McCormick's horse-drawn mechanical **reaper** quickly cut, or reaped, stalks of wheat from the field and gathered them up for processing. Farmers who used this new machine no longer had to harvest wheat by hand. The reaper helped increase wheat production throughout the Great Plains. It changed not only American farming but also many of the world's farming methods. McCormick's reaper was so successful that he opened a factory in Chicago to produce his machines, advancing both rural farming technology and urban manufacturing techniques.

HISTORICAL THINKING

1. **READING CHECK** What was significant about manufacturing with interchangeable parts?
2. **DRAW CONCLUSIONS** How did McCormick's mechanical reaper contribute to agricultural and urban growth?
3. **MAKE CONNECTIONS** McCormick's reaper connected both rural and urban innovation. How might another invention described in the text connect the country and the city?

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